

Spatial Statistics: Bayesian CAR Model of Obesity in Texas Counties

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1 Introduction

Obesity is one of the largest public health challenges in the United States, contributing to increased risk of diabetes, cardiovascular disease, and premature mortality. In Texas, obesity rates are among the highest in the nation, but prevalence varies across its 254 counties. Understanding the spatial distribution of obesity and identifying the socioeconomic factors that drive geographic variation is important for creating targeted public health interventions.

A key feature of health outcomes measured at the county level is that they tend to exhibit spatial autocorrelation. This means neighboring counties often share similar obesity rates due to shared economic conditions, cultural factors, food environments, and healthcare access. Standard regression models that ignore this spatial dependence produce biased standard errors and may lead to incorrect inferences. Spatial models which explicitly account for the correlation structure among neighboring areas provide more reliable estimates and are more accurate at quantifying uncertainty.

In this report, I analyzed the prevalence of adult obesity across all 254 Texas counties using data from the 2025 County Health Rankings (CHR), a program of the University of Wisconsin Population Health Institute. I fit a Bayesian hierarchical model with an improper conditional autoregressive (CAR) prior on spatial random effects using the CARBayes package in R. The covariates include the percentage of physically inactive adults, the percentage of adults with insufficient sleep, and median household income. Covariates were selected based on examining how each covariate is associated with the outcome, stepwise AIC-based selection, and variance inflation factor (VIF) diagnostics across all the predictors with no missing data (Appendix A.2). We assess spatial autocorrelation using Moran's I, evaluate convergence of the MCMC sampler, and compare the spatial model to a non-spatial baseline.

2 Data Description

The data for this analysis come from the 2025 County Health Rankings for Texas, downloaded from countyhealthrankings.org. The dataset provides health outcomes and socioeconomic indica-

tors for all 254 Texas counties. No observations are missing for any of the variables used in this analysis.

The response variable is the percentage of adults aged 20 and older with a body mass index (BMI) of 30 or greater, shown as **% Adults with Obesity**. Three covariates are included: **% Physically Inactive**, the percentage of adults who report no leisure time physical activity, **% Insufficient Sleep**, the percentage of adults who report sleeping fewer than 7 hours per night, and **Median Household Income**, measured in thousands of dollars. These covariates represent physical behavior, sleep behavior, and economic resources. They were selected from a pool of 12 predictors based on their predictive strength, low multicollinearity (all VIFs below 2.6), and interpretability.

Table 1: Descriptive statistics for the response variable and covariates across 254 Texas counties (2025 CHR).

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Obesity (%)	39.25	3.20	28.6	37.42	39.40	41.18	49.90
Inactivity (%)	29.98	5.18	17.4	26.90	29.45	32.75	47.30
Insuff. Sleep (%)	39.36	2.10	31.8	38.10	39.50	40.90	45.20
Income (\$1000s)	63.47	14.71	36.7	53.50	60.36	69.53	120.15

Table 1 shows descriptive statistics for all variables. The average county level obesity rate is 39.25% (SD = 3.2), ranging from 28.6% to 49.9%, indicating heterogeneity. Physical inactivity averages 29.98% (SD = 5.18), insufficient sleep averages 39.36% (SD = 2.1), and median household income averages \$63,467 (SD = \$14,709).

The correlation matrix reveals a strong positive correlation between obesity and physical inactivity ($r = 0.873$), a strong positive correlation between obesity and insufficient sleep ($r = 0.753$), and a moderate negative correlation between obesity and income ($r = -0.611$). Among the covariates, physical inactivity and insufficient sleep are positively correlated ($r = 0.716$), and both are negatively correlated with income ($r = -0.653$ and $r = -0.551$, respectively). Choropleth maps of the covariates (Appendix, Figure 3) confirm these patterns visually with physical inactivity mirroring the spatial distribution of obesity. Income is concentrated around metropolitan areas and insufficient sleep shows a more uniform distribution with less geographic variation.

The spatial structure of the data is defined by the county adjacency graph for Texas. Using counties that share a border or vertex are neighbors, the average number of neighbors per county is 5.7, with a range of 1 to 9. The adjacency matrix $\mathbf{W}$ is a 254×254 binary symmetric matrix where $W_{ij} = 1$ if counties i and j share a boundary or vertex, and 0 otherwise.

3 Methods

3.1 Spatial Autocorrelation Testing

Global spatial autocorrelation was assessed using Moran’s I statistic, computed on the outcome variable and on OLS residuals using the queen contiguity spatial weights matrix. A significant

Moran’s I indicates that neighboring counties tend to have similar values, motivating the use of spatial models.

3.2 OLS Baseline

An ordinary least squares (OLS) regression of obesity on physical inactivity, insufficient sleep, and median household income was fit as a non-spatial baseline. Moran’s I was then applied to the OLS residuals to test for residual spatial autocorrelation.

3.3 Bayesian CAR Model

We used a Bayesian hierarchical model with spatial random effects following the CAR framework. Let Y_i denote the obesity prevalence in county i ($i = 1, \dots, 254$). The data model is:

$$Y_i = \beta_0 + \beta_1 \times \text{Inactivity}_i + \beta_2 \times \text{Insuff_Sleep}_i + \beta_3 \times \text{Income}_i + \phi_i + \varepsilon_i$$

where ϕ_i is a spatial random effect and $\varepsilon_i \sim N(0, \nu^2)$ captures independent non-spatial variation. The spatial random effects $\phi = (\phi_1, \dots, \phi_{254})$ are assigned an improper CAR prior.

The model is implemented using the `S.CARleroux` function in the `CARBayes` R package with the adjacency matrix $\mathbf{W}$ constructed via queen contiguity from the U.S. Census Line county shapefiles. The Leroux CAR model provides a flexible framework that nests both the intrinsic CAR model ($\rho = 1$) and the independence model ($\rho = 0$) as special cases. When $\rho = 1$, the spatial random effects follow the intrinsic CAR specification, which assumes that the conditional mean of each random effect is the average of its neighbors. Posterior samples are obtained via the MCMC algorithm implemented in `CARBayes`.

3.4 Prior Distributions

We use the default `CARBayes` priors, which are diffuse and weakly informative:

- **Regression coefficients:** $\beta \sim N(0, 100,000)$, a vague normal prior centered at zero with very large variance. This prior is essentially flat over any plausible range of coefficient values, allowing the data to dominate.
- **Spatial variance:** $\tau^2 \sim \text{Inverse-Gamma}(1, 0.01)$, which has a prior mean of 0.01 and infinite variance.
- **Non-spatial variance:** $\nu^2 \sim \text{Inverse-Gamma}(1, 0.01)$, with the same specification.
- **Spatial autocorrelation:** $\rho = 1$ (fixed), corresponding to the improper CAR model.

Using weakly informative priors allow the data to dominate the posterior inference and sensitivity to the prior specification is minimal given the moderate sample size of 254 counties.

3.5 MCMC Inference

Posterior inference is conducted via MCMC simulations implemented in CARBayes. The MCMC sampler is run for a total of 110,000 iterations. The first 10,000 iterations are discarded as burn-in to allow the chain to reach its stationary distribution. The remaining 100,000 iterations are thinned by a factor of 10, keeping every 10th sample to reduce autocorrelation in the stored posterior samples. This yields 10,000 posterior samples for inference. The burn-in period of 10,000 iterations is conservative and was chosen after verifying that the trace plots show convergence before this point. The thinning factor of 10 was selected to balance storage efficiency while retaining the posterior information. The random seed is fixed for reproducibility.

Convergence is assessed using things like trace plots of all parameters which shows stationary behavior (Figure 2). Also, the Geweke diagnostics compares the mean of the first 10% and last 50% of the chain using a z -test and is computed for all parameters. Values between -2 and 2 indicate convergence. Lastly, effective sample sizes (ESS) is computed to assess the degree of autocorrelation in the chain.

3.6 Variable Selection

Covariates were selected from 12 candidate predictors with complete data across all 254 counties. The candidates included: % Physically Inactive, % Insufficient Sleep, Median Household Income, % Adults with Diabetes, % Adult Smoking, % Food Insecure, % Rural, % 65 and Over, % Hispanic, Income Inequality Ratio, % Children in Poverty, and % Uninsured Adults. The stepwise procedure selected Inactivity, Insufficient Sleep, Diabetes, and Income Ratio (Adj. $R^2 = 0.801$, AIC = 908.7), but this model exhibited multicollinearity (Inactivity VIF = 14.8, Diabetes VIF = 12.5). The final model (Inactivity + Insufficient Sleep + Income) was chosen for its strong fit (Adj. $R^2 = 0.794$, AIC = 916.9) with all VIFs below 2.6.

4 Results

4.1 Spatial Autocorrelation of Outcome

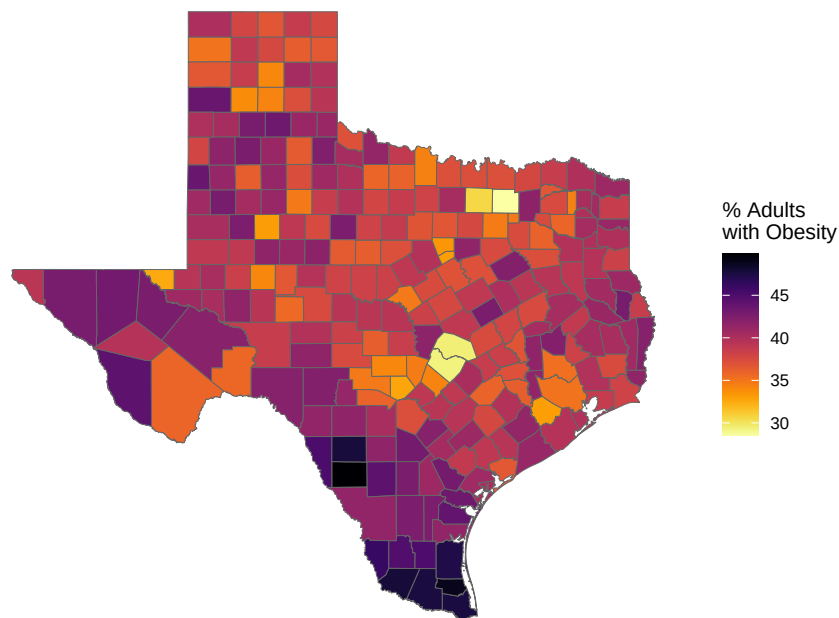
Before fitting the spatial model, we test for spatial autocorrelation using Moran's I. The global Moran's I statistic for obesity is 0.412 ($p < 2.2 \times 10^{-16}$), providing very strong evidence of positive spatial autocorrelation. This means that counties with high obesity rates tend to cluster near other high rate counties. Figure 1 shows the choropleth map of obesity prevalence, revealing a clear geographic pattern with higher rates concentrated in southern Texas and lower rates in northern and central-urban counties.

The covariates also exhibit spatial autocorrelation. Moran's I is 0.447 ($p < 2.2 \times 10^{-16}$) for physical inactivity, 0.254 ($p < 4e - 12$) for insufficient sleep, and 0.499 ($p < 2.2 \times 10^{-16}$) for income. The strong spatial clustering in income is consistent with the well-known geographic concentration of wealth in metropolitan areas versus rural counties.

4.2 OLS Baseline Results

An ordinary least squares (OLS) regression of obesity on the three covariates explains 79.62% of the variance. However, Moran’s I on the OLS residuals is 0.054 ($p = 0.062$), suggesting some residual spatial structure that the covariates alone do not fully capture. The DIC comparison (Section 4.4) confirms that the spatial model provides a meaningfully better fit, motivating its use.

Adult Obesity Prevalence by County, Texas (2025 CHR)



Source: County Health Rankings & Roadmaps, 2025

Figure 1: Adult obesity prevalence (%) by county, Texas, 2025 County Health Rankings.

4.3 Posterior Summaries

Table 2: Posterior summaries from the Bayesian improper CAR model. Asterisk (*) indicates the 95% credible interval excludes zero.

Parameter	Mean	SD	2.5%	Median	97.5%	Sig.
Intercept (β_0)	12.312	2.402	7.620	12.331	17.008	*
% Phys. Inactive (β_1)	0.381	0.036	0.309	0.382	0.449	*
% Insuff. Sleep (β_2)	0.410	0.069	0.275	0.409	0.547	*
Income, \$1000s (β_3)	-0.010	0.008	-0.026	-0.010	0.007	
τ^2 (spatial var.)	0.386	0.338	0.009	0.306	1.239	
ν^2 (non-spatial var.)	1.887	0.210	1.498	1.878	2.334	

Table 2 presents the posterior summaries for all model parameters. Physical inactivity is a strong predictor. A one percentage point increase in physical inactivity is associated with a 0.38 percent-

age point increase in obesity (95% CI: 0.31, 0.45), holding other covariates constant.

Insufficient sleep also shows a positive association with obesity with a one percentage point increase in the rate of insufficient sleep is associated with a 0.41 percentage point increase in obesity (95% CI: 0.27, 0.55). This is consistent with evidence linking short sleep duration to weight gain through hormonal disruption and increased caloric intake.

Median household income has a posterior mean of -0.01. Its 95% credible interval is (-0.026, 0.007). The credible interval includes zero, indicating that income does not have a statistically significant independent effect after accounting for the behavioral covariates and spatial structure.

The variance parameters give insight into the spatial structure. The non-spatial variance ν^2 has a posterior mean of 1.887, while the spatial variance τ^2 has a posterior mean of 0.386. The ratio $\tau^2/(\tau^2 + \nu^2) \approx 0.17$ indicates that after accounting for the covariates, roughly 17% of the remaining variation is spatially structured.

4.4 Model Comparison

To assess whether the spatial model improves upon a non-spatial model, we compare two models. The improper CAR model ($\rho = 1$) and a non-spatial model ($\rho = 0$). The CAR model has a DIC of 903.26, compared to 916.68 for the non-spatial model, a difference of 13.42 points. The WAIC shows a similar pattern (907.39 vs. 918.56) favoring the spatial model.

Also, Moran's I on the CAR model residuals is -0.035 ($p = 0.793$), indicating no remaining spatial autocorrelation. The spatial model has successfully captured the spatial structure in the data, whereas the OLS residuals still showed marginal spatial autocorrelation (Moran's I = 0.054, $p = 0.062$).

4.5 Convergence Diagnostics

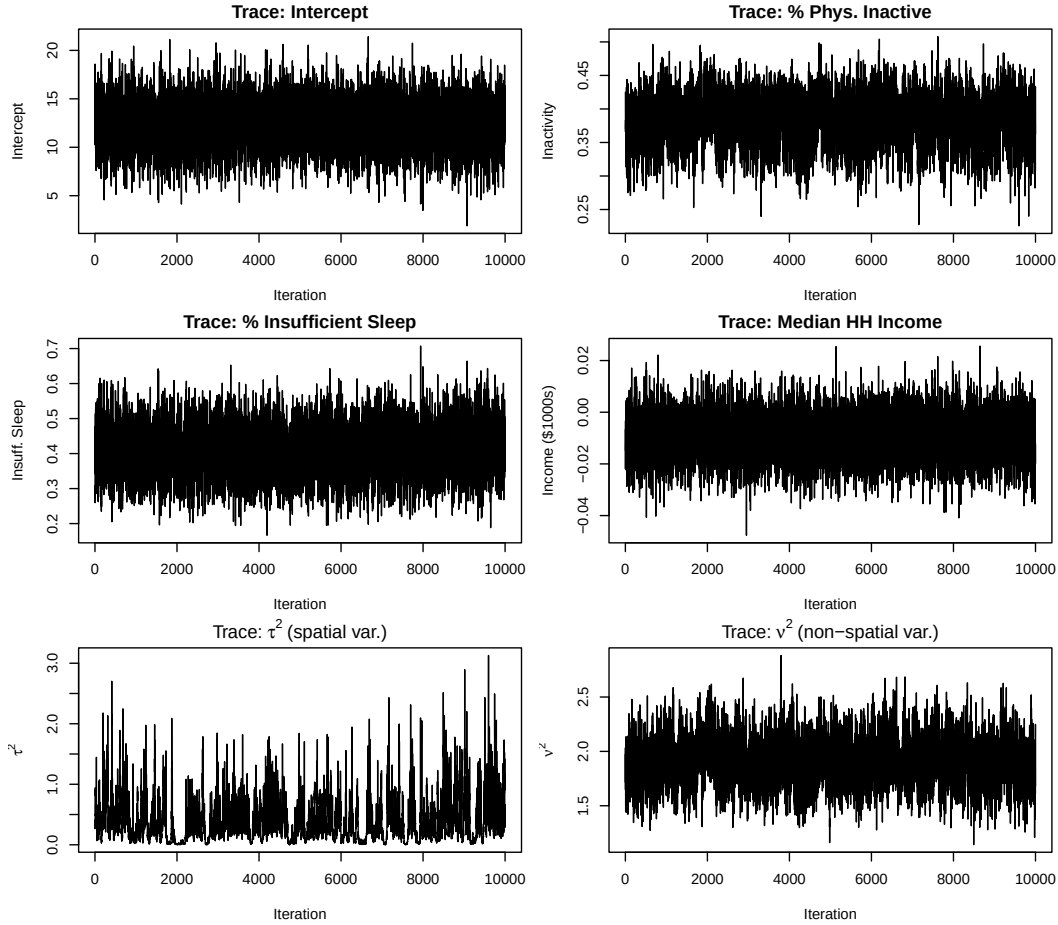


Figure 2: Trace plots of MCMC posterior samples for all model parameters (10,000 samples after burn-in and thinning).

Figure 2 shows the trace plots for all model parameters. The chains show good mixing with no visible trends, drifts, or multimodality. All Geweke diagnostic values fall within the convergence band: the intercept (0.22), physical inactivity (-1), insufficient sleep (0.57), income (-0.15), τ^2 (1.39), and ν^2 (-1.53). The effective sample sizes are adequate for all parameters with the regression coefficients ranging from 853 to 9603, while ν^2 has an ESS of 620 and τ^2 has an ESS of 321. Although the ESS for τ^2 is the lowest, it still provides a sufficient number of independent posterior draws for reliable inference.

5 Discussion and Conclusions

This analysis reveals that adult obesity prevalence in Texas exhibits significant spatial clustering (Moran's $I = 0.412$), and that a Bayesian CAR model with covariates provides a better fit than a standard non-spatial regression.

Physical inactivity is a dominant predictor of county level obesity, with a posterior mean effect of 0.38 percentage points per one percentage point increase in inactivity. This is a strong positive association and is consistent with the historic link between sedentary behavior and weight gain, and is robust to the inclusion of spatial random effects.

Insufficient sleep is also a meaningful predictor, with a posterior mean effect of 0.41 percentage points per one percentage point increase in insufficient sleep prevalence. This aligns with evidence linking short sleep duration to weight gain. Counties where adults are more sleep deprived tend to have higher obesity rates, even after controlling for physical inactivity and income.

The spatial random effects capture residual geographic variation not explained by the covariates. Maps of posterior median spatial effects (Appendix) reveal that counties along the South Texas border have notably higher than expected obesity rates even after controlling for inactivity, sleep, and income, while counties in the north show slightly lower than expected rates. Most of the state is near zero, indicating the covariates explain the bulk of the spatial variation. These residual patterns may reflect unmeasured factors such as lack of access to healthy food, cultural dietary differences, or access to recreational activities.

However, there are some limitations. For example the CHR data relies on model based small area estimation for some counties, which introduces measurement uncertainty not accounted for in our model. Also, the set of covariates is limited with future analysis that could include other influential covariates. The improper CAR model also assumes a single level of spatial smoothing across all counties. Models that allow for spatially varying coefficients or boundary detection could capture more complex patterns. Finally, insufficient sleep and physical inactivity may share common causes such as long work hours making it difficult to fully remove their independent contributions.

In conclusion, this analysis demonstrates the importance of spatial statistical methods for understanding geographic variation in health outcomes. The Bayesian CAR model successfully removed residual spatial autocorrelation and identified physical inactivity and insufficient sleep as the primary behavioral drivers of county level obesity in Texas. These results suggest that public health resources should be directed toward promoting both physical activity and healthy sleep habits in the high prevalence counties of Texas.

6 References

County Health Rankings & Roadmaps (2025). University of Wisconsin Population Health Institute. *2025 County Health Rankings Texas Data*. Available at: <https://www.countyhealthrankings.org>.

Appendix

A.1 Additional Figures

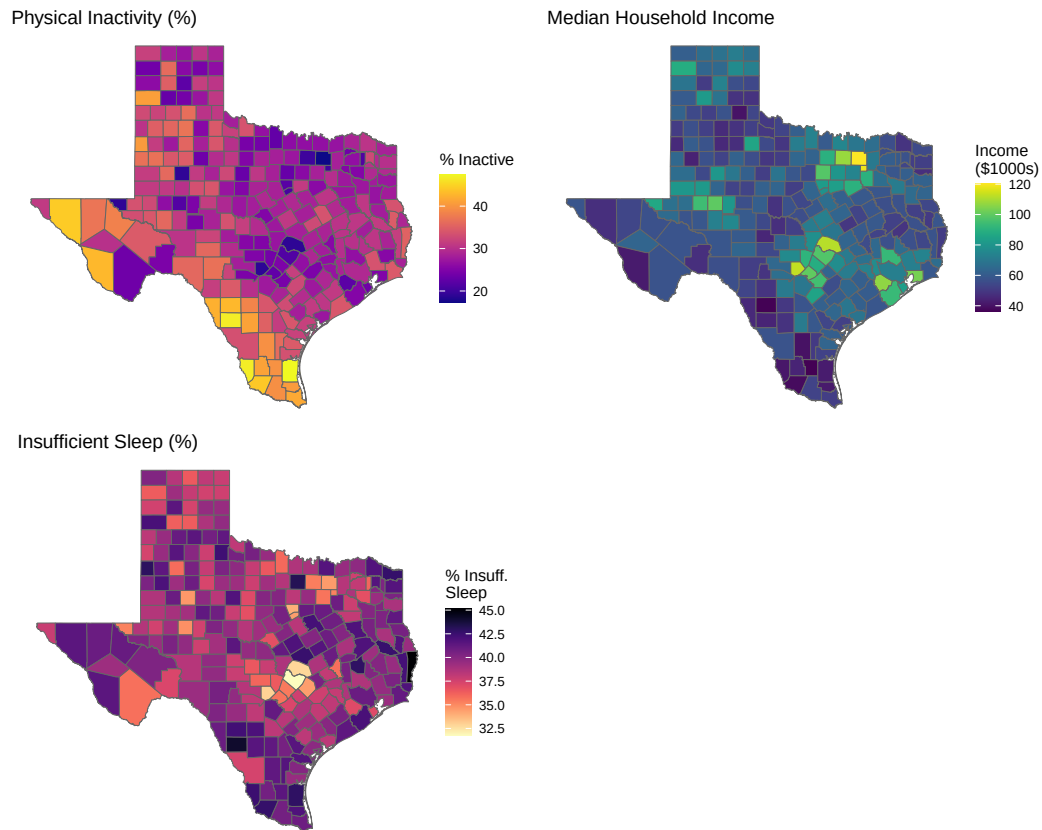


Figure 3: Choropleth maps of covariates across Texas counties.

Posterior Median of Spatial Random Effects

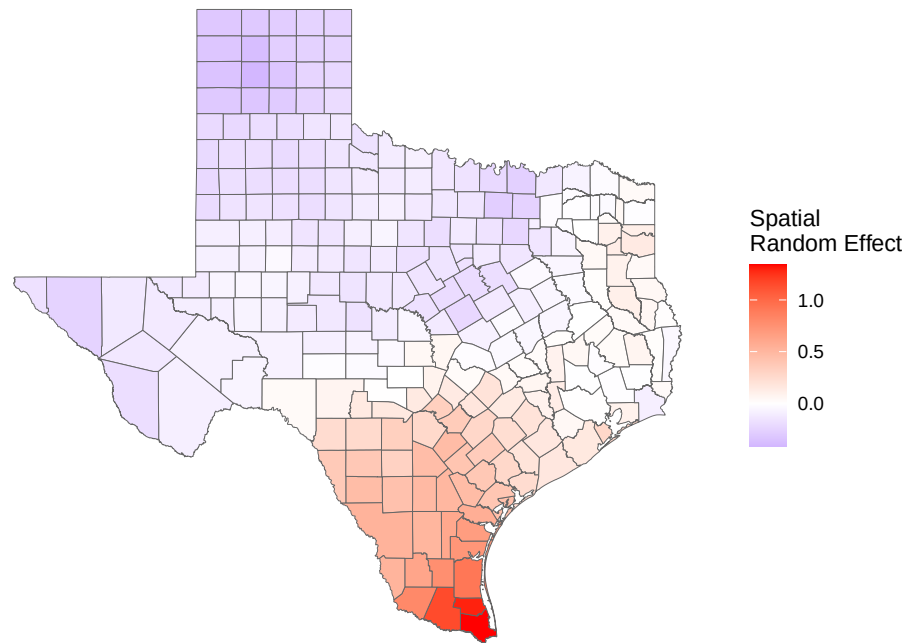
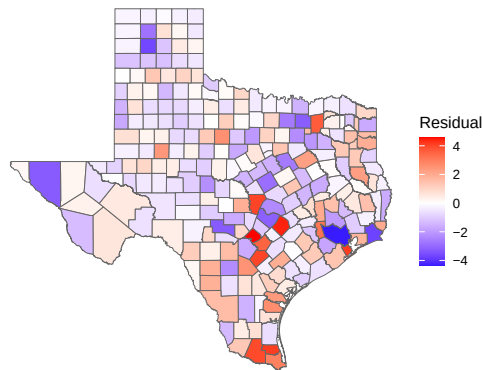


Figure 4: Posterior median of spatial random effects. Red indicates higher-than-expected obesity; blue indicates lower-than-expected.

OLS Residuals



CAR Model Residuals

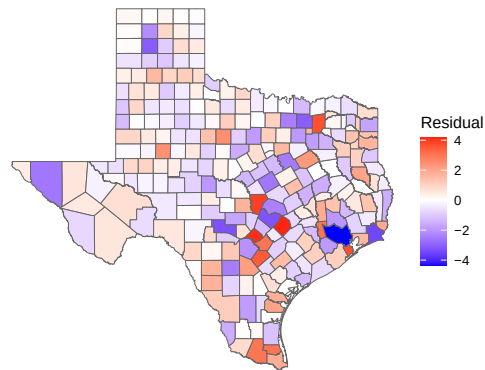


Figure 5: Comparison of OLS residuals (left) and CAR model residuals (right). The CAR model eliminates residual spatial patterns.

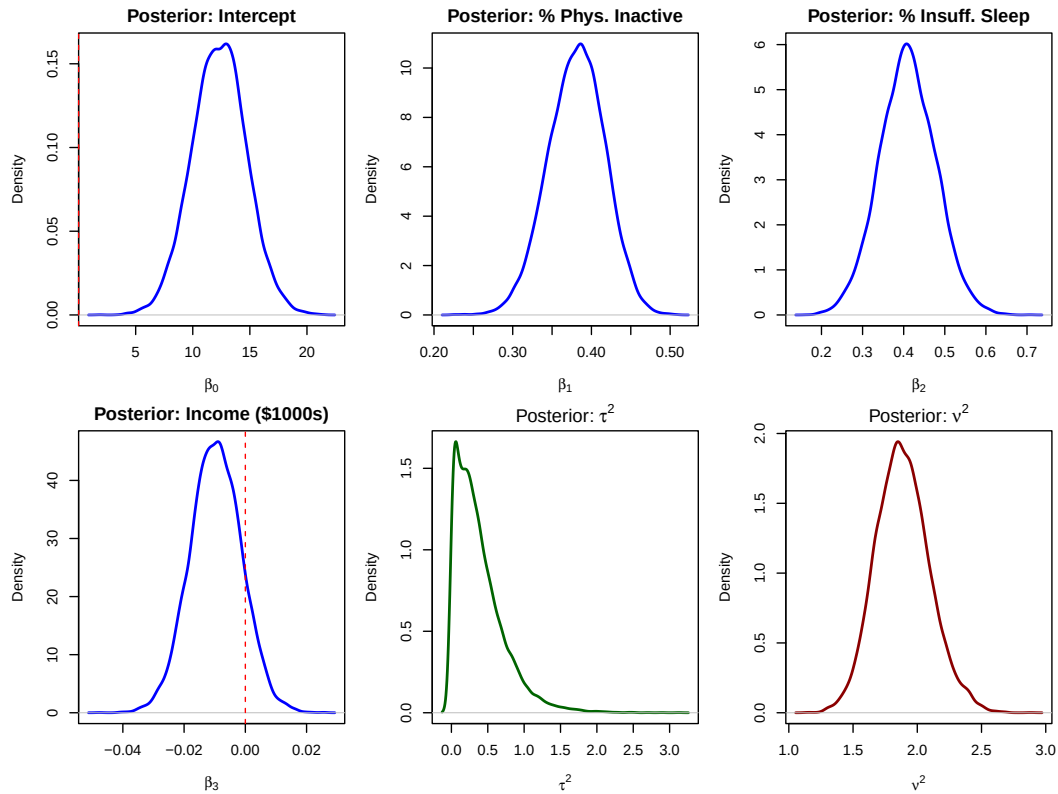


Figure 6: Marginal posterior density plots for all model parameters. The dashed red line marks zero.

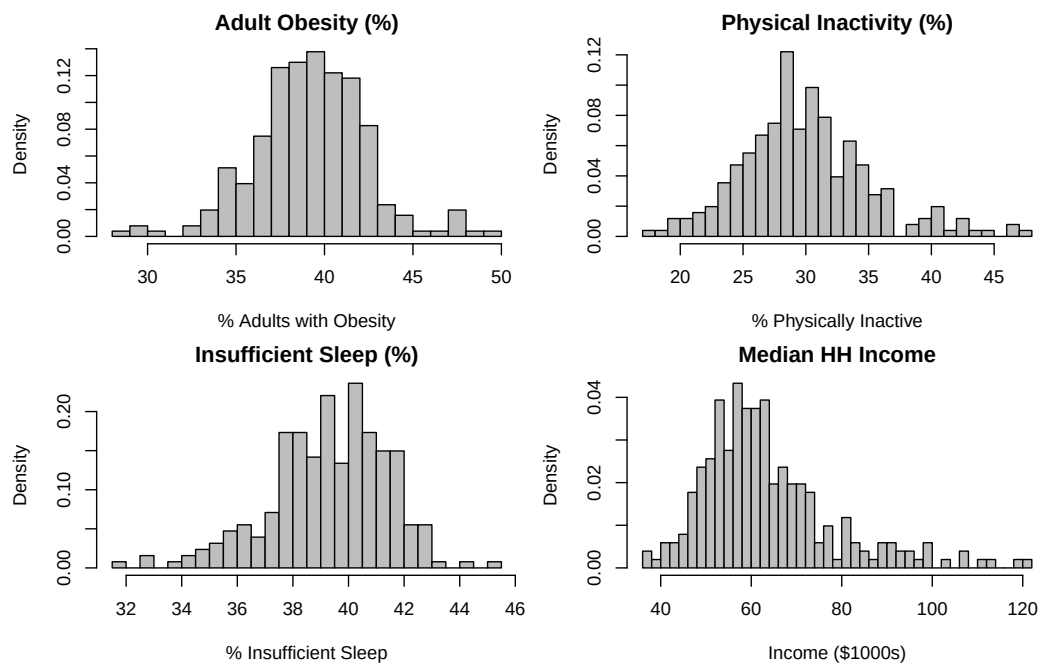


Figure 7: Histograms of all variables used in the analysis.

A.2 Variable Selection Notes

Covariates were selected from 12 candidate predictors with complete data across all 254 counties. The candidates included: % Physically Inactive, % Insufficient Sleep, Median Household Income, % Adults with Diabetes, % Adult Smoking, % Food Insecure, % Rural, % 65 and Over, % Hispanic, Income Inequality Ratio, % Children in Poverty, and % Uninsured Adults. The stepwise procedure selected Inactivity, Insufficient Sleep, Diabetes, and Income Ratio (Adj. $R^2 = 0.801$, AIC = 908.7), but this model exhibited multicollinearity (Inactivity VIF = 14.8, Diabetes VIF = 12.5). The final model (Inactivity + Insufficient Sleep + Income) was chosen for its strong fit (Adj. $R^2 = 0.794$, AIC = 916.9) with all VIFs below 2.6.

A.3 R Code for Model Fitting

```
library(readxl)
library(sf)
library(spdep)
library(CARBayes)
library(coda)
library(dplyr)
library(tigris)

# Read data
raw_add <- read_excel("2025 County Health Rankings Texas Data - v3.xlsx",
                      sheet = "Additional Measure Data", col_names = FALSE,

health_data <- data.frame(
  FIPS = as.character(raw_add[[1]]),
  County = raw_add[[3]],
  Obesity = as.numeric(raw_add[[111]]),
  Inactivity = as.numeric(raw_add[[215]]),
  Insuff_Sleep = as.numeric(raw_add[[150]]),
  Income = as.numeric(raw_add[[267]])
)
health_data <- health_data[!is.na(health_data$County), ]
health_data$FIPS <- sprintf("%05s", health_data$FIPS)
health_data$Income_k <- health_data$Income / 1000

# Spatial setup
tx_counties <- counties(state = "TX", year = 2022, cb = TRUE)
tx_counties$GEOID <- as.character(tx_counties$GEOID)
tx_map <- merge(tx_counties, health_data, by.x = "GEOID", by.y = "FIPS")

tx_nb <- poly2nb(tx_map, queen = TRUE)
```

```

tx_weights <- nb2WB(tx_nb)
N <- nrow(tx_map)

# Adjacency matrix
rep_tx <- rep(1:N, tx_weights$num)
W <- matrix(0, N, N)
for (i in 1:N) {
  W[i, tx_weights$adj[rep_tx == i]] <- rep(1, tx_weights$num[i])
}

# Moran's I
tx_listw <- nb2listw(tx_nb)
moran.test(tx_map$Obesity, tx_listw, rank = TRUE)

# OLS baseline
lm_obesity <- lm(Obesity ~ Inactivity + Insuff_Sleep + Income_k,
                 data = st_drop_geometry(tx_map))
moran.test(residuals(lm_obesity), tx_listw, rank = TRUE)

# Bayesian improper CAR model
model_data <- st_drop_geometry(tx_map) %>%
  select(Obesity, Inactivity, Insuff_Sleep, Income_k)

set.seed(295)
model_car <- S.CARleroux(
  formula = Obesity ~ Inactivity + Insuff_Sleep + Income_k,
  data = model_data,
  family = "gaussian",
  W = W,
  rho = 1,
  burnin = 10000,
  n.sample = 110000,
  thin = 10,
  prior.mean.beta = NULL,
  prior.var.beta = NULL,
  prior.nu2 = NULL,
  prior.tau2 = NULL,
  verbose = TRUE
)

print(model_car$summary.results)
print(model_car$modelfit)

# Non-spatial comparison

```

```
model_nospatial <- S.CARleroux(  
  formula = Obesity ~ Inactivity + Insuff_Sleep + Income_k,  
  data = model_data, family = "gaussian", W = W, rho = 0,  
  burnin = 10000, n.sample = 110000, thin = 10, verbose = FALSE  
)
```